NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 24/03/2026
Time: 3 Hrs | GTM – 03 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
D	D	B	C	C	A	C	B	B	A
11	12	13	14	15	16	17	18	19	20
A	A	C	A	C	B	C	B	B	A
21	22	23	24	25					
2	400	20	8	600					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
B	A	C	C	C	C	A	D	A	C
36	37	38	39	40	41	42	43	44	45
C	C	D	C	A	C	D	C	C	B
46	47	48	49	50					
1787	974	3	4	68					

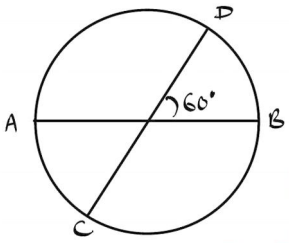
MATHEMATICS

51	52	53	54	55	56	57	58	59	60
A	D	C	A	B	A	D	A	D	D
61	62	63	64	65	66	67	68	69	70
B	A	B	C	D	C	D	A	D	D
71	72	73	74	75					
4	5	198	2	58					

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HINTS
PHYSICS

1.



$$V_A = -\frac{kp}{r^2}$$

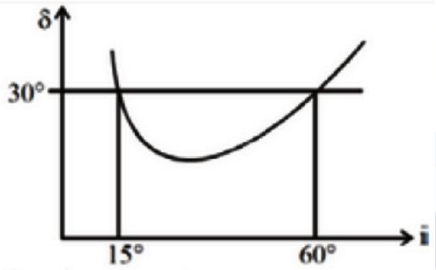
$$V_B = \frac{kp}{r^2}$$

$$V_C = -\frac{kp\cos 60^\circ}{r^2}$$

$$V_D = \frac{kp\cos 60^\circ}{r^2}$$

2. $\mu_d = \frac{v_d}{E} = \frac{\sigma}{ne}$ independent of current and cross section area

3.



$$d = i + e - A$$

$$30^\circ = 15^\circ + 60^\circ - A$$

$$A = 45^\circ$$

For same deviation, i & e can b interchanged

4. Any point on the equatorial line of an electric dipole is equidistant from the positive and the negative charges on the dipole.

The net electric potential at the point is,

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$$V = \frac{kq}{r} + \frac{k(-q)}{r} = 0$$

5. $\frac{(2n+1)v}{4l} = 500$

$\frac{(2n+3)v}{8l} = 350$

$\Rightarrow \frac{2n+1}{2n+3} = \frac{250}{350} = \frac{5}{7}$

$\Rightarrow n = 2\left(5^{th}\text{ Harmunic }\right)$

$\frac{5\times 340}{4\times l} = 500 \Rightarrow l = 85\text{ cm.}$

6. Let $z = \exp\left[-(x - vt + 2)^2\right]$

Put $z = \exp\left[-(x - v + 2)^2\right]$

$\Rightarrow \quad x - v + 2 = x - 2$

$\Rightarrow \quad v = +4m/s$

7. $Y = \frac{W/A}{\Delta l/l}$

$\frac{W}{\Delta l} = \frac{YA}{l}$

Slope $\propto A$

8. $\lambda_1 T_1 = \lambda_2 T_2$

$\frac{T_1}{T_2} = \frac{\lambda_2}{\lambda_1} = 2$

Rate of heat loss $\dot{Q} = 4\pi r^2 \sigma e T^4$

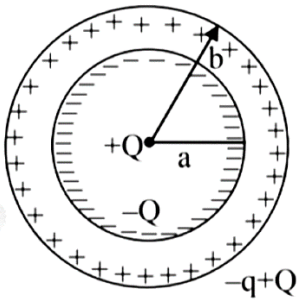
$\frac{\dot{Q}_1}{\dot{Q}_2} = \left(\frac{r_1}{r_2}\right)^2 \left(\frac{T_1}{T_2}\right)^4 = 4$

$\dot{Q} = -ms\frac{d\theta}{dt} = -\frac{4}{3}\pi r^3 \rho s \frac{d\theta}{dt}$

$\frac{\left(\frac{d\theta}{dt}\right)_1}{\left(\frac{d\theta}{dt}\right)_2} = \left(\frac{\dot{Q}_1}{\dot{Q}_2}\right) \left(\frac{r_2}{r_1}\right)^3 = 32$

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9.



To make E.F. zero inside the conductor, By gauss law, Charge –Q(equal and opposite to that of central charge) comes on the inner surface of the shell.

So, by charge conservation on the shell, outer surface charge –q+Q

10. $\frac{F}{a}+\rho gh=\frac{\mu_g}{A}$

$F=37.5N$

Correct option (A)

11. The current will flow through R_L when the diode is forward biased.

12. Least count of measuring instrument = 0.1 s

Mean absolute error = $\frac{\sum\left|t_{avg}-t_i\right|}{3}$ = 0.17 ≈ 0.2 s

13. $\frac{v_1\sin\theta_1}{g}-\frac{v_2\sin\theta_2}{g}=T$

$\frac{v_1^2\sin^2\theta_1}{2g}-\frac{v_2^2\sin^2\theta_2}{2g}=H$

$\Rightarrow\ m(v_1\sin\theta_1+v_2\sin\theta_2)=\frac{2mH}{T}$

14. $W=$ area under $p-V$ graph

$A_2>A_1>A_3$

$\therefore\ W_2>W_1>W_2$

15. Temperature becomes 4 times

PV^2 = const

$P\propto T^2$

$P_2 = 16P_1$; $V_2=\frac{V_1}{4}$

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Molar heat capacity $C = C_V + \frac{R}{1-m} = \frac{3}{2}R - R = \frac{R}{2}$

Heat given = $\frac{1}{2} \times \left(4P_1V_1 - P_1V_1\right) = 1.5P_1V_1$

16. $mgh + W_f = \frac{1}{2}mv^2 f - 0$

Or $1 \times 10 \times 1 + W_f = \frac{1}{2} \times 1 \times 2^2 - 0$

$\therefore W_f = -8J$

17. (C) Conceptual

18. $U + K = 2$

For maximum K, U should be minimum, and so $U + K = 2$ or $\frac{d}{dx}\left(\frac{x^4}{4} - \frac{x^2}{2}\right) = 0$

Or $x = 0, \pm 1$

$U_{\min} = \frac{(1)^4}{4} - \frac{(1)^2}{2} = \frac{1}{4}J$

$\therefore -\frac{1}{4} + \frac{1}{2} \times 1 \times v_{\max}^2 = 2$

Or $v_{\min} = \frac{3}{\sqrt{2}} ms^{-1}$

19. At steady state the current in $1k\Omega$ to $2k\Omega$ resistor is $i = \frac{6}{3} = 2mA$

The potential difference across $1\mu F$ capacitor $V_1 = iR = 2 \times 2 = 4$ *volt*, and charge on $1\mu F$, $q_1 = C, V_1 = 1 \times 4 = 4\mu C$, The potential difference across $2\mu F$ capacitor is also, $V_2 = 4$ volt and so charge on $2\mu F$ capacitor is

$q_2 = C_2V_2 = 2 \times 4 = 8\mu C$

20. Here, $u = f + x_1, v = f + x_2$

Use $f = \frac{uv}{u+v}$ and solve to get $f = \sqrt{x_1x_2}$

21. Given $V = 7V$, $r = 5\Omega$

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$$R_{eq} = \frac{40\times120}{40+120}\Omega$$

$$I = \frac{V}{R} = \frac{7}{5+\frac{40\times120}{40+120}} = \frac{7}{5+30} = \frac{1}{5} = 0.2A$$

22. $2\mu_0 t = n \times 10000 = (n + 1) \times 5000$ (where μ_0 is 1.25)

$\therefore$ n = 1

$2\mu_0 t = 1 \times 10000 \times 10^{-10}$

t = 0.4 μm

23. $4(1 + 20 \times 10^{-6}\,\Delta T) = 3.998\,(1 + 10 \times 10^{-6}\,\Delta T)$

$\Delta T = -50$ °C

common temperature is 30 –50 = –20° C

24. $E_1 = 13.6$ eV

$E_n = -\frac{13.6eV}{n^2}$, $KE_n = \frac{13.6eV}{n^2} = \frac{1}{2}I\omega^2$ .....(i)

and $I\omega = \frac{nh}{2\pi}$

$\therefore$ from (i)

$$\frac{E_1}{n^2} = \frac{1}{2}\left(\frac{nh}{2\pi}\right)\omega$$

$$f = \frac{2E_1}{n^3 h}$$

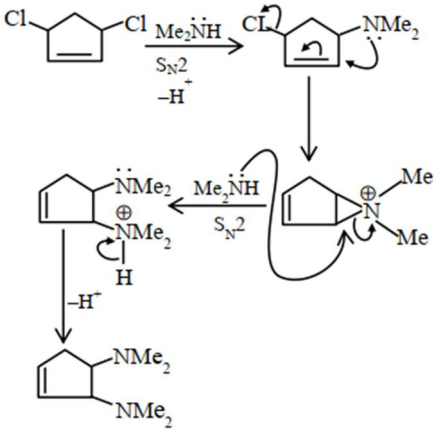
25. $$F = 0.2\times\frac{P}{hc/\lambda}\times\frac{2h}{\lambda} + 0.5\frac{P}{hc/\lambda}\times\frac{h}{\lambda} = \frac{0.9P}{C}$$

$$= \frac{0.9\times200}{3\times10^8} = 6\times10^{-7}\ N$$

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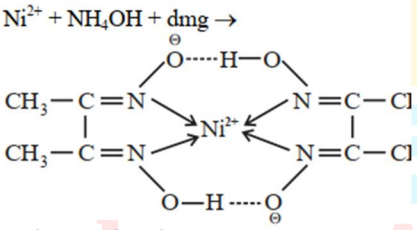
CHEMISTRY

26.



The above mechanism valid for both cis and trans isomers. So the products are same for both cis and trans isomers.

27.



2 Five member ring

III II

$Fe_4[Fe(CN)_6]_3$

Prussian Blue

28.

Step1 : Dissociation

$$Al(OH)_3 \rightleftharpoons Al^{3+} + 3OH^-$$

Step2 : Concentrations

If solubility = S mol/L,

then $[Al^{3+}] = S, [OH^-] = 3S.$

Step3 : ExpressionofKsp

$$K_{sp} = [Al^{3+}][OH^-]^3 = S \cdot (3S)^3 = 27S^4$$

Step4 : Convertgivensolubility

(Wg / 100mL)intomol / L

$$S = \frac{W}{M} \times \frac{1000}{100} = 10\frac{W}{M}$$

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Step5 : SubstituteintoKsp $K_{sp} = 27(10\frac{W}{M})^4$

$$K_{sp} = 27\times10^4\left(\frac{W}{M}\right)^4 \quad K_{sp} \approx 2.7\times10^5\left(\frac{W}{M}\right)^4$$

Final

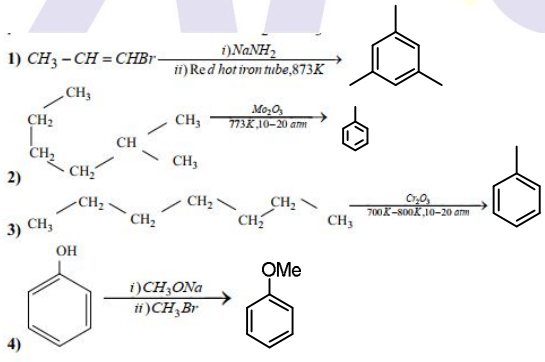
Answer : $(A)\ 10^5\left(\frac{W}{M}\right)^4$

29. P) CH_3CHO vs $PhCHO$ → I (Iodoform test)
Acetaldehyde gives iodoform; benzaldehyde doesn't.
Q) CH_3CH_2OH vs CH_3OCH_3 → I, IV
Ethanol gives iodoform (+) and is a primary alcohol in Victor–Meyer; ether doesn't.
R) Cane sugar vs Lactose → II (Barfoed test)
Lactose is reducing (reacts), cane sugar is non-reducing (no reaction).
S) $CH_3CH_2NH_2$ vs $PhNH_2$ → V (Azo-dye test)
Aniline (aromatic 1° amine) forms azo dye; ethylamine doesn't.

30. Rf depends on solvent polarity → Statement-1 is false.
TLC is adsorption chromatography → Statement-2 is true.
Answer: (3) Statement-2 is true but Statement-1 is false.

31. Reason (short):
Only a phenol dissolves in aq. NaOH → B must be cresol (Ar–OH).
Remaining isomers with formula C_7H_8O are benzyl alcohol (Ph–CH_2OH) and anisole (Ph–OCH_3).
Boiling points: benzyl alcohol > cresol > anisole (H-bonding > phenol > ether).
So A = Ph–CH_2OH, B = cresol (CH_3–C_6H_4–OH), C = anisole (Ph–OCH_3) → Option 3.

32. $NaNH_2$ eliminates HBr from CH_3–CH=CHBr → CH_3–C≡CH (propyne); on red-hot Fe tube (873 K) propyne trimerizes to 1,3,5-trimethylbenzene (mesitylene).
Mesitylene has three identical o/p-directing Me groups at 1,3,5; all ring positions available for EAS are equivalent, so mono-chlorination with $Cl_2/FeCl_3$ gives only one product.



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33. according to slow step $r(t) = K_2[N_2O_2][H_2]$

Where $K_1 = \frac{[N_2O_2]}{[NO]^2}$

$[N_2O_2] = K_1[NO]^2$

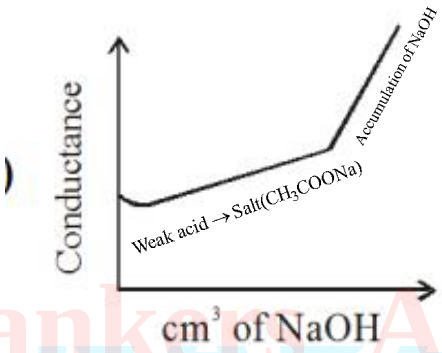
So, $r(t) = K_2K_1[NO]^2[H_2]$

So, $r(t) = K_2K_1[NO]^2[H_2] = K[NO]^2[H_2]$

$-\frac{1}{2}\frac{d[H_2]}{dt} = -\frac{1}{2}\frac{d[NO]}{dt} = \frac{d[N_2]}{dt}$

Where $K = K_1K_2$

34. Weak acid (CH_3COOH) has low initial conductance.
Adding NaOH forms CH_3COO^-/Na^+ → conductance rises slowly.
After equivalence, excess OH^- from strong base → sharp rise.



35. Correct Answer = (C) Ammoniacal $AgNO_3$

Reagent	Effect on Isopropyl alcohol (2°)	Effect on t-Butyl alcohol (3°)	Distinguishes?
$KMnO_4$	Oxidized to acetone	No oxidation	Yes
HCI + anhy. $ZnCl_2$ (Lucas test)	Reacts slowly - alkyl chloride forms	Reacts instantly → t-BuCI forms	Yes
Ammoniacal $AgNO_3$	No characteristic reaction (used for alkyl halides, not alcohols)	No reaction	No
$K_2Cr_2O_7$	Oxidized to ketone	No oxidation	Yes

36. Typical field strengths: CN^- strong, H_2O weak, NH_3 (with Co^{3+}) effectively strong/low spin, F^- weak

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Complex	Metal ox. state → d^n	Field	CF filling (octahedral)	Unpaired e^-
$[Fe(CN)_6]^{4-}$	$Fe^{2+} \rightarrow d^6$	strong	$t_{2g}^6 e_g^0$ (low spin)	0
$[Fe(H_2O)_6]^{2+}$	$Fe^{2+} \rightarrow d^6$	weak	$t_{2g}^4 e_g^2$ (high spin)	4
$[Co(NH_3)_6]^{3+}$	$Co^{3+} \rightarrow d^6$	strong (low spin with NH_3)	$t_{2g}^6 e_g^0$	0
$[MnF_6]^{3-}$	$Mn^{3+} \rightarrow d^4$	weak	$t_{2g}^3 e_g^1$ (high spin)	4

37. In Graph (I) - two radial nodes are present
$n - l - 1 = 2$ $n = 5$ for $l = 2$
In Graph (II) one radial node is present
$n - l - 1 = 2$ $n = 4$ for $l = 2$
In Graph (III) zero radial node
$n - l - 1 = 2$ $n = 3$ for $l = 2$
Graph (IV) does not represents any radial distribution curve

38. (1) $2CH_3–CO–CH_3 + Ba(OH)_2$ (aldol condensation under mild conditions)
→ Forms diacetone alcohol $(CH_3)_2C(OH)–CH_2–COCH_3$.
Correct.
(2) $2CH_3–CO–CH_3 + HCl$ (strong acid, dehydration)
→ Gives mesityl oxide $(CH_3)_2C=CH–COCH_3$.
Correct.
(3) $2CH_3–CO–CH_3$ → condensation + further aldol with CH_3COCH_3
→ Final product is phorone: $(CH_3)_2C=CH–CO–CH= C(CH_3)_2$.
Correct.
(4) $CH_3CHO + CH_3COCH_3 + NaOH$ → cross aldol condensation
Enolate of acetone attacks acetaldehyde → product is $CH_3–C(OH)–CH_2–CH_2–COCH_3$.
That compound is 4-hydroxyhexan-2-one, not 4-hydroxypentan-2-one (one carbon more).
Incorrect match.

39. (A) Sc > Y > La (Ionisation energy)
Down group 3 (Sc, Y, La), atomic size ↑ → IE decreases.
So: Sc > Y > La : Correct.
(B) $Hf^{2+} < Hf \simeq Zr$ (Size)
Hf and Zr have similar atomic radii due to lanthanide contraction.
But Hf^{2+} cation must be smaller than neutral Hf.
So order should be $Hf^{2+} < Hf \simeq Zr$: Correct.
(C) Al > Si (Electronegativity)
Electronegativity: Al ≈ 1.5, Si ≈ 1.8.
So Si > Al, not Al > Si. : Incorrect.
(D) Ga > Al (Ionisation energy)
Normally down group IE decreases, but Ga has slightly higher IE than Al due to poor

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shielding by 3d electrons.

So Ga > Al : Correct.

40.

Moles of $H_2SO_4 = M \times V = 1 \times 0.100 = 0.10$ mol

Mass of initial solution $= 1.5 \times 100 = 150$ g

Mass of water added $= 400 \times 1 = 400$ g

Total mass of solution $= 150 + 400 = 550$ g

$$\text{Final volume} = \frac{550}{1.25} = 440 \text{ mL} = 0.440 \text{ L}$$

$$\text{Final molarity} = \frac{0.10}{0.440} \approx 0.227 \text{ M} \approx 0.23 \text{ M}$$

41. $A \rightarrow B + C + D + E + \ldots + x + y + z$

$A \rightarrow B + C + D + E + \ldots\ldots\ldots + x + y + z$

$t = 0$	P_0	0	0	0	0		$0+0+0$	
$t = t$	$P_0 - x$	x	x	x	x		x	x x
$t = \infty$	0	P_0	P_0	P_0	P_0		P_0	P_0 P_0

$\therefore$ At t = t:

$P_0 + 24x = P$

$$x = \frac{P - P_0}{24}$$

And

$$P_0 - x = P_0 - \frac{P - P_0}{24} = P_0 - \frac{P}{24} + \frac{P_0}{24} = \frac{25P_0 - P}{24}$$

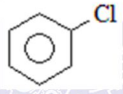
$$K = \frac{1}{t}\ln\left(\frac{24P_0}{25P_0 - P}\right) = \frac{2.303}{t}\log\left(\frac{24P_0}{25P_0 - P}\right)$$

42. For $n = 4, m = -3$:

The only possible value of l is 3.

Therefore, the orbital angular momentum is:

$$\sqrt{l(l+1)}\frac{h}{2\pi} = \sqrt{3(3+1)}\frac{h}{2\pi} = \frac{2\sqrt{3}h}{2\pi} = \frac{\sqrt{3}h}{\pi}$$

43. A is  where Cl is ring deactivating & less reactive.

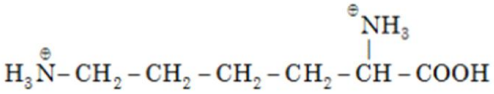
44. Reason: NaOBr does Hofmann bromamide reaction → only primary amides ($-CONH_2$) convert to amines with loss of one carbon.

(A) secondary amide ✗, (B) tertiary amide ✗, (D) nitrile ✗, (C) aryl primary amide ✓ → p-chloroaniline.

45. CoS (group 4) > CuS(group 2)

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47.

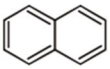


Basic amino acid Isoelectric point = $\frac{8.95+10.53}{2} = 9.74$

$x = 9.74 \Rightarrow 100x = 974$

48.

(1)  (2) 

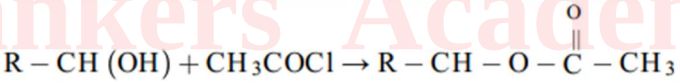
(3)  (4) 

(5)  (6) 

(7)  (8) 

(3), (7) and (8) are aromatic.

49.



For every acetylation reaction, molecular mass increases by 42

No. of OH group $= \frac{318-150}{42} = 4$

50. % of sulphur = $\frac{\text{weight of sulphur}}{\text{weight of org.comp}} \times 100$

$\frac{0.32}{0.47} \times 100 = 68\%$

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MATHEMATICS

51. Solution Equation of the parabola is

$$\left(y+\frac{35a}{16}\right)=\frac{a^3}{3}\left(x+\frac{3}{4a}\right)^2 \text{ whose vertex is}$$

$$x=-\frac{3}{4a}, y=-\frac{35a}{16} \Rightarrow xy=\frac{105}{64}$$

52. using Telescopic Mathod.

53. equation of circle is $x^2+y^2-\frac{4}{5}x-\frac{44}{5}y+\frac{84}{5}=0$

54. $x^3+x-1=9 \Rightarrow x=2$

$\Rightarrow g(9)=2$

$g(f(x))=x \Rightarrow g'\left(f(x)\cdot f'(x)\right)=1$

$$\Rightarrow g'(f(x))=\frac{1}{f'(x)}=\frac{1}{f'(2)}=\frac{1}{13}$$

Put $f(x)=9$

$\Rightarrow x=2$

$$\Rightarrow \frac{g(9)}{g'(9)}=26$$

55. $4\cos^2\theta-1=4\left(1-\sin^2\theta\right)-1=3-4\sin^2\theta=\frac{\sin 3\theta}{\sin\theta}$ …(i)

So, given expression can be written as

$$=\frac{\sin 27^\circ}{\sin 9^\circ}\times\frac{\sin 81^\circ}{\sin 27^\circ}\times\frac{\sin 243^\circ}{\sin 81^\circ}\times\frac{\sin 729^\circ}{\sin 243^\circ} \quad [\because \text{By(i)}]$$

$$=\frac{\sin 729^\circ}{\sin 9^\circ}=1$$

56. $Q=PAP^T$

$P^TQP=A\left(\because PP^T=I\right)$

$P^TQ^{2005}P=AP^TQ^{2004}P$

$=A^2P^TQ^{2003}P=A^3P^TQ^{2002}P=\ldots.$

$=A^{2004}P^T(QP)=A^{2004}P^T(PA)$

$$\left(\because Q=PAP^T \Rightarrow QP=PA\right)=A^{2005}=\begin{bmatrix}1 & 2005\\ 0 & 1\end{bmatrix}$$

57. Let the chord of contact is PQ which touches $x^2+4y^2=4$ at R

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Now, assume $R = (2\cos\theta,\ 2\sin\theta)$

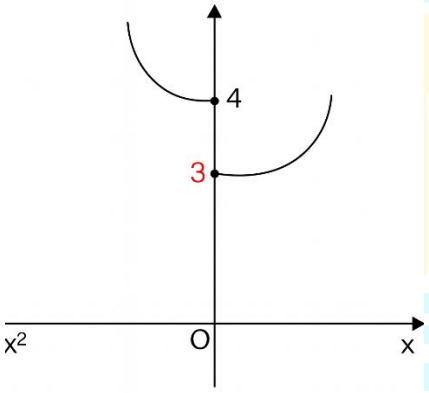
Equation of the chord of contact PQ is $hx + 2yk = 6\ldots\ldots(i)$

Again, the equation of the tangent PQ is $\frac{x\cos\theta}{2} + \frac{y\sin\theta}{2} = 1\ldots.(ii)$

From (i) and (ii), we get $\frac{h2}{\cos\theta} = \frac{2k}{\sin\theta} = 6 \Rightarrow \cos\theta = \frac{h}{3}, \sin\theta = \frac{k}{3} \Rightarrow x^2 + y^2 = 9$

Hence, the perimeter of the circle is 6π units

58. $^6C_2(\sqrt{x})^4\left(\frac{\sin x}{x}\right)^2 = \frac{15}{4}\sin^2 x = \frac{1}{4}$



$\Rightarrow \lim\limits_{x\to 0^-} f(\cos^3 x - \cos^2 x) + \lim\limits_{x\to 0^-} f(\sin^2 x - \sin^3 x)$

$= f(0^-) + f(0^+)$

$= 7$

60. Put $x = 1 \Rightarrow g(1) = 2$ Diff. $6g(x) = 3\left(xg'(x) + g(x)\right) - 3x^2$

$\frac{xg'(x) - g(x)}{x^2} = 1 \frac{d}{dx}\left(\frac{g(x)}{x}\right) = 1$

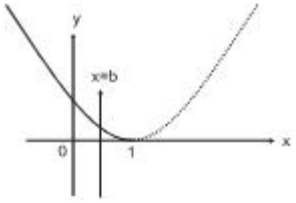
$\frac{g(x)}{x} = x + c \Rightarrow g(x) = x(x+1)$

$g(2) = 2(3) = 6$

61. $R_1 = \int_0^b (x-1)^2 dx = \frac{(x-1)^3}{3}\Big|_0^b = \frac{(b-1)^3 + 1}{3}$

also $R_2 = \int_b^1 (x-1)^2 dx = \frac{(x-1)^3}{3}\Big|_b^1 = -\frac{(b-1)^3}{3}$

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$$R_1 - R_2 = \frac{2(b-1)^3}{3} + \frac{1}{3}$$

$$\Rightarrow \frac{1}{4} = \frac{2(b-1)^3}{3} + \frac{1}{3} \Rightarrow (b-1)^3 = -\frac{1}{8}$$

$$b = \frac{1}{2}$$

62. Let r be the radius of spherical balloon

S = Surface area

$S = 4\pi r^2$

$$\frac{dS}{dt} = 8\pi r \times \frac{dr}{dt} = k \quad \text{(constant)}$$

$4\pi r^2 = kt + C$ (C is constant of integration)

For $t = 0, r = 3 \Rightarrow 36\pi = C$

For $t = 5, r = 7 \Rightarrow K = 32\pi$

$4\pi r^2 = 32\pi t + 36\pi$

$r^2 = 8t + 9$

for t = 9

$r^2 = 81$

r = 9

63. $$\sum_{i=1}^{n} a_i = \frac{n}{2}\{2a_1 + (n+1)\} = 192 \Rightarrow 2a_1 + (n-1) = \frac{384}{n}$$

$$\sum_{i=1}^{n/2} a_{2i} = \frac{n}{4}\left[2a_1 + 2 + \left(\frac{n}{2} - 1\right)2\right] = 120$$

$$2a_1 + n = \frac{480}{n}$$

From equation (2) and (1)

$$1 = \frac{480}{n} - \frac{384}{n}$$

$n = 480 - 384 = 96$

64. range of f(x) is [3, 13]

for g(x) to be continuous

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$$\frac{f(x)}{k.\text{sgn}\left(x^2+1\right)}=\frac{f(x)}{k}<1$$

$\Rightarrow k>f(x)$

$\Rightarrow k>13\Rightarrow k=14$

65. $|A| = 3$

$$\left|\text{adj}\left(-4\text{adj}\left(-3\text{adj}\left(3\text{adj}\left((2A)^{-1}\right)\right)\right)\right)\right|$$

$$\left|-4\text{adj}\left(-3\text{adj}(3\text{adj}(2A)^{-1}\right)\right|^2$$

$$4^6\left|\text{adj}\left(-3\text{adj}\left(3\text{adj}(2A)^{-1}\right)\right)\right|^2$$

$$2^{12}\cdot 3^{12}\left|3\text{adj}(2A)^{-1}\right|^8$$

$$2^{12}\cdot 3^{12}\cdot 3^{24}\left|\text{adj}(2A)^{-1}\right|^8$$

$$2^{12}\cdot 3^{36}\left|(2A)^{-1}\right|^{16}$$

$$2^{12}\cdot 3^{36}\frac{1}{|2A|^{16}}$$

66. Let the co-ordinates of A be $(3\lambda+3, 8-\lambda, \lambda+3)$ and the co-ordinates of B be $(-3\mu-3, 2\mu-7, 4\mu+6)$.

Then direction ratios of AB are $3\lambda+3\mu+6,\ -\lambda-2\mu+15,\ \lambda-4\mu-3$

$AB \perp L_1$ so $3(3\lambda+3\mu+6)-(-\lambda-2\mu+15)+(\lambda-4\mu-3)=0$

i.e. $11\lambda+7\mu=0$

and $AB \perp L_2$ so $-3(3\lambda+3\mu+6)+2(-\lambda-2\mu+15)+4(\lambda-4\mu-3)=0$

i.e. $-7\lambda-29\mu=0$

$\Rightarrow \lambda=\mu=0=0$ so the point A is (3, 8, 3) and the point B is (–3, –7, 6)

$AB=\sqrt{36+225+9}=\sqrt{270}=3\sqrt{30}$

67. By substituting $x=\frac{1}{t}$ and $dx=-\frac{1}{t^2}dt$ in equation (1),

$$I=-\int_2^{1/2}\frac{\tan^{-1}\frac{1}{t}}{\frac{1}{t}}\cdot\frac{1}{t^2}\cdot dt$$

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$$I = -\int_2^{1/2} \frac{\tan^{-1}\frac{1}{t}}{t}dt = \int_{1/2}^{2} \frac{\cot^{-1}t}{t}dt = \int_{1/2}^{2} \frac{\cot^{-1}x}{x}dx \ldots.(2)$$

On adding equation (1) and (2), we have

$$2I = \int_{1/2}^{2} \frac{\tan^{-1}x + \cot^{-1}x}{x}dx; 2I = \frac{\pi}{2}\int_{1/2}^{2}\frac{dx}{x} = \frac{\pi}{2}[\log x]_{1/2}^{2}$$

$$2I = \frac{\pi}{2}\left[\log_e 2 - \log_e \frac{1}{2}\right] = \pi \log_e 2$$

68. $A = \{x \in \mathbb{R} : [x+3] + [x+4] \le 1\}$

$2[x] + 7 \le 1$

$2[x] \le -6$

$[x] \le -3 \Rightarrow x < -2$ …(A)

$$B = \left\{x \in R : 3^x \left(\sum_{r=1}^{\infty}\frac{3}{10^r}\right)^{x-3} < 3^{-3x}\right\}$$

$$3^x \left(\sum_{r=1}^{\infty}\frac{3}{10^r}\right)^{x-3} < 3^{-3x}$$

$$3^{2x-3}\left(\frac{\frac{1}{10}}{1-\frac{1}{10}}\right)^{x-3} < 3^{-3x}$$

$$\Rightarrow \left(\frac{1}{9}\right)^{x-3} < 3^{-5x+3}$$

$\Rightarrow 3^{6-2x} < 3^{3-5x}$

$\Rightarrow 6 - 2x < 3 - 5x \Rightarrow 3x < -3 \Rightarrow x < -1$

69. $\frac{dy}{dx} - y \tan x = 2x \sec x$

$y(0) = 0$

I.F. $= e^{-\int \tan x\, dx} = e^{-\log \sec x}$

I.F. $= \cos x$

$\cos x \cdot y = \int 2x \sec x . \cos x\, dx$

$\cos x . y = x^2 + c$

$c = 0$

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$y = x^2 \sec x$

$y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{16}.\sqrt{2} = \frac{\pi^2}{8\sqrt{2}}$

$y'\left(\frac{\pi}{4}\right) = \frac{\pi}{2}.\sqrt{2} + \frac{\pi^2}{16}.\sqrt{2}$

$y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{9}.2 = \frac{2\pi^2}{9}$

$y'\left(\frac{\pi}{3}\right) = 2\frac{\pi}{2}.2 + \frac{\pi^2}{9}.2.\sqrt{3}$

$\frac{4\pi}{3} + \frac{2\pi^2\sqrt{3}}{9}$

70. Sum of roots < 1

$\Rightarrow \lambda^2 - 5\lambda + 5 < 1 \Rightarrow (\lambda - 1)(\lambda - 4) < 0 \Rightarrow$

$1 < \lambda < 4$ ...(1)

Product of roots < 1

$\Rightarrow 2\lambda^2 - 3\lambda - 5 < 0 \Rightarrow (2\lambda - 5)(\lambda + 1) < 0 \Rightarrow -1 < \lambda < \frac{5}{2}$ ...(2)

(1) & (2) $\Rightarrow 1 < \lambda < \frac{5}{2}$.

71. $\text{Lim}_{x\to 0} \frac{\int_0^x \left(t^2 + e^{t^2}\right)^{\frac{1}{1-\cos t}} dt}{\left(e^x - 1\right)}$

using Leibnitz Rule

$\lim_{x\to 0} \frac{\left(x^2 + e^{x^2}\right)^{\frac{1}{2\sin^2\frac{x}{2}}}}{e^x} = e^{\lambda}$

$\lambda = \lim_{x\to 0} \frac{x^2 + e^{x^2} - 1}{2\sin^2\frac{x}{2}} = \frac{x^2}{2\sin^2\frac{x}{2}} - \frac{1 - e^{x^2}}{2\sin^2\frac{x}{2}} = 4$

72. $p(A \cap \bar{B}) + p(\bar{A} \cap B) = \frac{5}{9}$

$p(A)p(\bar{B}) + p(\bar{A})p(B) = \frac{5}{9}$

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$\therefore p(1-2p)+(1-p)2p=\frac{5}{9}$ $\Rightarrow 9p(1-2p)+18p(1-p)=5$

$\Rightarrow 9p-18p^2+18p-18p^2=5$ $\Rightarrow 36^2p-27p+5=0$

$\Rightarrow (3p-1)(12p-5)=0$ $p=\frac{1}{3}$ or $p=\frac{5}{12}$

$p_{max}=\frac{5}{12}=\lambda$

73. $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}\begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$

$a^2+b^2+c^2+d^2+e^2+f^2+g^2+h^2+i^2=5$

Case -I : Five (1's) and four (0's)

${}^9C_5=126$

Case- II : One ,(2) and one (1)

${}^9C_2\times 2!=72$ $\therefore$ Total $=198$

74. $\Delta=\begin{vmatrix} 4 & k & 2 \\ k & 4 & 1 \\ 2 & 2 & 1 \end{vmatrix}=0 \Rightarrow 8-k(k-2)-2(2k-8)=0$

$\Rightarrow 8-k^2+2k-4k+16=0 \Rightarrow -k^2-2k+24=0$

$\Rightarrow k^2+2k-24=0 \Rightarrow (k+6)(k-4)=0 \Rightarrow k=-6,4$

Number of values of k is 2

75. SMALL

A _ _ _ _ # $\frac{4!}{2!}=12$

L _ _ _ _ # $4!=24$

M _ _ _ _ # $\frac{4!}{2!}=12$

SA _ _ _ # $\frac{3!}{2!}=3$

SL _ _ _ # $3!=6$

S M A L L # 1

58th position

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